ABSTRACT

➢ **Project Title:** Cancer Detection System using Deep Learning

➢ **Group Members:**

Aryan Jadhav

Ganraj Kendre

Varnan Dondage

Soham Bhosage

➢ **Background & Brief Preview:**

Cancer is one of the leading problems in the world today. There are about 18.1 million cancer patients in the world today. Cancer can be difficult to detect in the early stages. We can use Artificial Intelligence to find out patterns in the reports of patients in the early stages and help in the prevention of cancerous tumors from growing any further. Artificial intelligence is a computer program or algorithm that uses past data to make future predictions. It teaches itself how to analyze and interpret data, which helps machine learning algorithms may pick up on patterns that are not readily discernable to the human eye or brain. As these algorithms are exposed to more new data, their ability to learn and interpret the data improves. Researchers have also used deep learning, a type of machine learning, in cancer imaging applications. Deep learning refers to algorithms that classify information in ways much as the human brain does. Deep learning tools use "artificial neural networks" that mimic how our brain cells take in, process, and react to signals from the rest of our body. Research on AI for cancer imaging: Studies suggest that AI has the potential to improve the speed, accuracy, and reliability with which doctors answer those questions. But what scientists are most excited about is the potential for AI to go beyond what humans can currently do themselves. AI can "see" things that we humans can't, and can find complex patterns and relationships between very different kinds of data.

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- **Technology:** Python

- **Application**:

Cancer Detection is an application software that Radiologists can use to detect if a tumor is malicious or benign. This means that the software can predict if the tumor is cancerous or not on the basis of the tumor report. This software takes all the details of cancer, like the size, depth, and color of the tumor, and then finds similar patterns from past data. This helps the program figure out the gravity of cancer. Which tells the radiologists to either remove the tumor or not.

Signs of Committee Members after Approval:

Signs of Students: